KENDRIYA VIDYALAYA SANGATHAN,BHUBANESWAR REGION
SESSION:2024-25
PRE BOARD-I (MARKING SCHEME)
MAHEMATICS-BASIC(241)

SECTION-A

1.	(D) xy^2	1
2.	(C) ∠ B= ∠ D	1
3.	(B)56 cm	1
4.	(B) 6 cm	1
5.	(A)1.5	1
6.	(D) 12:1	1
7.	(B) $-\frac{3}{2}$	1
8.	(D) $\frac{12}{8}$	1
9.	(B)) $\frac{AD}{BD} = \frac{AE}{EC}$	1
10.	(A) – 8	1
11.	(B) 30	1
12.	(C) $\tan 90^0$	1
13.	(B) $x = 100^0$, $y = 85^0$	1
14.	(D)Both Intersecting and coincident	1
15.	(B)499	1
16.	(A) 1 , 1	1
17.	(C)2520	1
18.	(B) 70°	1
19.	(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).	1
20.	(d) Assertion (A) is false but Reason (R) is true.	1
	SECTION-B	
21.	Substituting the correct value of T-ratios - Finding answer = $\frac{7}{12}$	1 1

<table>
<tr><td>22.</td><td>$PA^2 = PB^2$
$\Rightarrow (x-4)^2 + (y-3)^2 = (x-3)^2 + (y-4)^2$
$\Rightarrow x = y \quad or \quad x - y = 0$

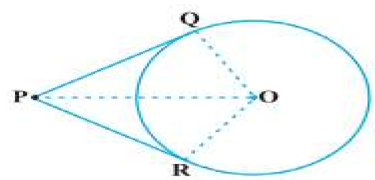
OR

$x = \dfrac{2.3 + 3 \cdot (-2)}{2+3} = 0$

$y = \dfrac{2 \cdot (-5) + 3.5}{2+3} = 1$</td><td>1
1

1

1</td></tr>
<tr><td>23</td><td>Correct figure
Given, To prove

Correct proof
</td><td>½
½

1</td></tr>
<tr><td>24</td><td>$\frac{12}{2}[2 \times 20 + 11d]$=900
$\Rightarrow d = 10$
Also $a_{12} = 20 + 11 \times 10 = 130$

OR

We know that
$a_n = a + (n-1)d$
$a_3 = a + (3-1)d = a + 2d = 5$
and $a_7 = a + (7-1)d = a + 6d = 9$
Solving the pair of linear equations (1) and (2), we get
$a = 3, \quad d = 1$
Hence, the required AP is 3, 4, 5, 6, 7, . . .</td><td>½
½
1

½
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½</td></tr>
<tr><td>25</td><td><table><tr><td>CI</td><td>No of students</td><td>Cumulative frequency</td></tr><tr><td>10-25</td><td>2</td><td>2</td></tr><tr><td>25-40</td><td>3</td><td>5</td></tr><tr><td>40-55</td><td>7</td><td>12</td></tr><tr><td>55-70</td><td>6</td><td>18</td></tr><tr><td>70-85</td><td>6</td><td>24</td></tr><tr><td>85-100</td><td>6</td><td>30</td></tr></table>
Finding correct value, n/2 =15 , median class-55-70 , l=55 , cf=12 ,f=6, h=15
Correct formula
Median=62.5</td><td>½

½

1/2
½</td></tr>
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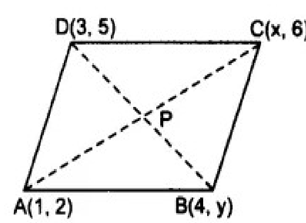
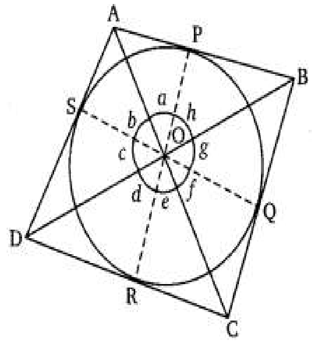
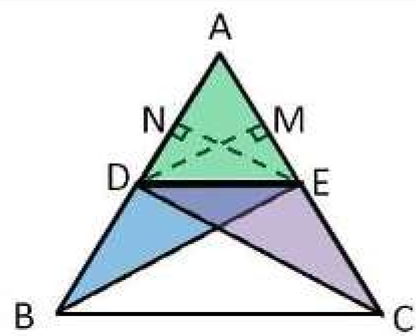
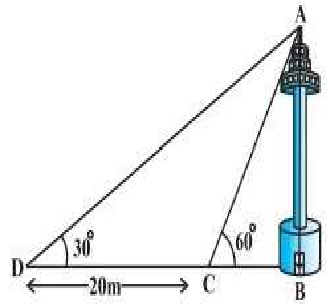
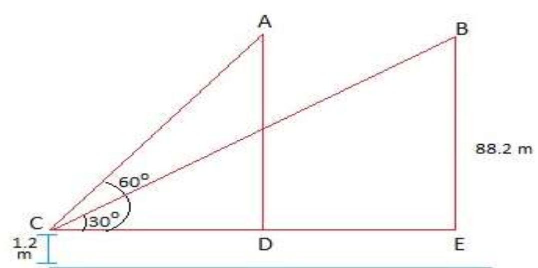
26	**Sol.** Let $\sqrt{5}$ is a rational number and $\sqrt{5} = \frac{a}{b}$, where a and b are coprime and $b \neq 0$. Now, $(\sqrt{5})^2 = \left(\frac{a}{b}\right)^2$ $\Rightarrow \quad 5b^2 = a^2 \quad \ldots(i)$ $\Rightarrow$ 5 is a factor of a^2 $\therefore$ a is also divisible by 5. Let $a = 5c$, where c is some integer. Substituting $a = 5c$ in (i) we get $5b^2 = (5c)^2 \Rightarrow 5b^2 = 25c^2 \Rightarrow b^2 = 5c^2$ $\Rightarrow$ 5 is a factor of b^2 $\therefore$ 5 is a factor of b. $\therefore$ 5 is a common factor of a and b This contradicts the fact that a and b are coprime so, our assumption is wrong. Hence, $\sqrt{5}$ is irrational.	1 1 1
27	We have, $(\text{cosec}\,\theta - \cot\theta)^2 = \frac{1-\cos\theta}{1+\cos\theta}$ LHS $= (\text{cosec}\,\theta - \cot\theta)^2$ $= \left(\frac{1}{\sin\theta} - \frac{\cos\theta}{\sin\theta}\right)^2 = \left(\frac{1-\cos\theta}{\sin\theta}\right)^2$ $= \frac{(1-\cos\theta)^2}{\sin^2\theta} = \frac{(1-\cos\theta)^2}{1-\cos^2\theta}$ $= \frac{(1-\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)} = \frac{1-\cos\theta}{1+\cos\theta}$ = RHS. Hence, **proved**.	1 1 1
28	Mid-point of AC = Mid-point of BD $\Rightarrow \quad \frac{x+1}{2}, \frac{6+2}{2} = \frac{4+3}{2}, \frac{y+5}{2}$ $\Rightarrow \quad \frac{x+1}{2} = \frac{7}{2}$ and $\frac{6+2}{2} = \frac{y+5}{2}$ $\Rightarrow \quad x + 1 = 7$ and $8 = y + 5$ $\Rightarrow \quad x = 7 - 1$ and $y = 8 - 5 = 3$ $\Rightarrow \quad x = 6$ and $y = 3$  OR	½ 1 ½ 1

	Given the points are (–2, 2) and (7, –4) Let the points of trisection be A and B Using the section formula we have, (–2, 2) —1— A —2— (7, –4) Coordinates of A are as: $\left[\left(\frac{1(7)+2(-2)}{1+2}\right),\left(\frac{1(-4)+2(2)}{1+2}\right)\right]$ $=\left[\left(\frac{3}{3}\right),\left(\frac{0}{3}\right)\right]$ $=(1,0)$ Now, the coordinates of B are as: (–2, 2) —2— B —1— (7, –4) $\left[\left(\frac{2(7)+1(-2)}{2+1}\right),\left(\frac{2(-4)+1(2)}{2+1}\right)\right]$ $=\left[\left(\frac{12}{3}\right),\left(\frac{-6}{3}\right)\right]$ $=(4,-2)$ Hence, the coordinates of points of trisection are (1, 0) and (4, –2).	½ 1 ½ 1
29	Solve x + 2y = 3 and write it as x = 3 – 2y Substitute the value of x . We get 7(3 – 2y) – 15y = 2 i.e., 21 – 14y – 15y = 2 i.e., – 29y = –19 Therefore, y = 19 /29 Substituting this value of y , , we get x= 49/ 29 Therefore, the solution is x = 49/29 , y = 19/29 .	1 1 1
30	Choosing an appropriate method Correct tabulation Correct formula FINDING CORRECT MEAN = **152.89**	1 ½ ½ 1
31	Let the quadrilateral ABCD touches the circle at P, Q, R,S(CORRECT FIGURE) AP=AS ------(i) (Tangents drawn from an external point to a circle are equal in BP=BQ -----(ii) length) CR=CQ -----(iii) DR=DS ------(iv) On adding (i), (ii), (iii) and (iv) AB+CD = AD + BC. (Hence Proved) OR	1 1 1

	In the figure, P, Q, R and S are the points touching the circle and sides AB, BC, CD and DA of the quadrilateral ABCD respectively.  From the figure, we observe that OA bisects $\angle SOP$. So, $\angle a = \angle b$ ... (i) Similarly, $\angle c = \angle d$ ... (ii) $\angle e = \angle f$ ... (iii) $\angle g = \angle h$ ... (iv) $\therefore$ $2(\angle a + \angle h + \angle e + \angle d) = 360°$ $\Rightarrow$ $(\angle a + \angle h) + (\angle e + \angle d) = 180°$ $\Rightarrow$ $\angle AOB + \angle DOC = 180°$. Similarly, $\angle AOD + \angle BOC = 180°$ Thus, opposite sides of quadrilateral ABCD subtend supplementary angles at the centre of a circle. Hence, Proved.	1 1 1
32	Correct figure Given ,to prove, construction Correct proof  	1 1 3
33	. Let us say, the two consecutive positive integers be x and x + 1. Therefore, as per the given statement, $x^2 + (x + 1)^2 = 365$ $\Rightarrow x^2 + x^2 + 1 + 2x = 365$ $\Rightarrow 2x^2 + 2x – 364 = 0$ $\Rightarrow x^2 + x – 182 = 0$ $\Rightarrow x^2 + 14x – 13x – 182 = 0$ $\Rightarrow x(x + 14) -13(x + 14) = 0$	1 1 1

	⇒ (x + 14)(x – 13) = 0 Thus, either, x + 14 = 0 or x – 13 = 0, ⇒ x = – 14 or x = 13 since, the integers are positive, so x can be 13, only. So, x + 1 = 13 + 1 = 14 Therefore, the two consecutive positive integers will be 13 and 14. OR Let the speed of the stream be x km/hr. Distance upstream = Distance downstream = 24 km Speed of the boat going upstream=18 - x Speed of the boat going downstream=18+x Time taken going upstream = 24/18-x Time taken going downstream = 24/18+x Now as per question $\frac{24}{18-x}-\frac{24}{18+x}=1$ $\Rightarrow\frac{432+24x+432+24x}{(18-x)(18+x)}=1$ $\Rightarrow x^2+48x-324=0$ $\Rightarrow x^2+54x-6x-324-2$ ⇒x(x+54)−6(x+54)=0 ⇒(x+54)(x−6)=0 ⇒x+54=0orx−6=0 ⇒x=−54orx=6 ∴ x=6 (Speed of the stream cannot be negative) Thus, the speed of stream is 6 km/hr.	1 1 1 1 1 1 1
34	As the horse is tied at one end of a square field, it will graze only a quarter (i.e. sector with θ = 90°) of the field with a radius 5 m. Here, the length of the rope will be the radius of the circle, i.e. r = 5 m It is also known that the side of the square field = 15 m **(i)** Area of circle = πr^2 = 22/7 × 5^2 = 78.5 m^2 Now, the area of the part of the field where the horse can graze = ¼ (the area of the circle) = 78.5/4 = **19.625** m^2 **(ii)** If the rope is increased to 10 m, Area of circle will be = πr^2 =22/7×10^2 = 314 m^2 Now, the area of the part of the field where the horse can graze = ¼ (the area of the circle) = 314/4 = 78.5 m^2 ∴ increase in the grazing area = 78.5 m^2 – 19.625 m^2 = **58.875** m^2	1 1 1 1 1

35	Given, AB is the height of the tower. DC = 20 m (given) As per given diagram, In right ΔABD, tan 30° = AB/BD 1/√3 = AB/(20+BC) AB = (20+BC)/√3 … (i)  Again, In right ΔABC, tan 60° = AB/BC √3 = AB/BC AB = √3 BC … (ii) From equation (i) and (ii) √3 BC = (20+BC)/√3 3 BC = 20 + BC 2 BC = 20 BC = 10 Putting the value of BC in equation (ii) AB = 10√3 So, **the height of the tower is 10√3 m and the width of the canal is 10 m.** **OR** Let the initial position of the balloon be A and final position be B. Height of balloon above the girl height = 88.2 m – 1.2 m = 87 m. **To Find**: Distance travelled by the balloon = DE = CE – CD  In right ΔBEC, tan 30° = BE/CE 1/√3= 87/CE CE = 87√3 In right ΔADC, tan 60° = AD/CD √3= 87/CD CD = 87/√3 = 29√3 DE = CE – CD = (87√3 – 29√3) = 29√3(3 – 1) = 58√3 Distance travelled by the balloon = **58√3 m.**	 1 ½ 1 ½ 1 1 ½ ½ 1 ½ 1 ½ 1
	SECTION-E	
36	(i) n(E)={(5,5) , (4,6) , (6,4)} =3 $P(E) = \frac{n(E)}{n(S)} = \frac{\mathbf{3}}{\mathbf{36}} = \frac{\mathbf{1}}{\mathbf{12}}$ (ii) P(Getting same number on two dice) = $\frac{6}{36} = \frac{1}{6}$ P(Not Getting same number on two dice)= 1- P(Getting same number on two dice) = $\mathbf{1} - \frac{\mathbf{1}}{\mathbf{6}} = \frac{\mathbf{5}}{\mathbf{6}}$ OR	½ ½ 1 1

	$n(E)=\{(2,2)(2,3),(2,5),(3,2),(3,3),(3,5),(5,2),(5,3),(5,5)\} =9$ P(Imran will get prime numbers on both the dice)= $\frac{\mathbf{9}}{\mathbf{36}}=\frac{\mathbf{1}}{\mathbf{4}}$ (iii) (1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5). n(getting odd numbers on both dice).=9 P(getting odd numbers on both dice) = $\frac{\mathbf{9}}{\mathbf{36}}=\frac{\mathbf{1}}{\mathbf{4}}$	1 1 ½ ½
37	(i) The height of the cylindrical part = 5 cm–(1.4+1.4) cm =**2.2** cm (ii) the total volume of one gulab jamun = Volume of cylinder + Volume of two hemispheres $= \pi r^2 h+(4/3)\pi r^3$ $= 4.312\pi+(10.976/3)\pi$ $= 25.05$ cm^3 Amount of sugar syrup in 1 gulab jamun= 30% of 25.05 = 7.515 cm^3 So ,total sugar syrup in 45 gulab jamuns= 45 X 7.515= **338.175 cm**^{3} OR space will be occupied by all the 45 gulab jamuns = Volume of 45 gulab jamuns $= \pi r^2 h+(4/3)\pi r^3 \times 45$ $= 4.312\pi+(10.976/3)\pi \times 45$ $= 25.05$ cm^3 X 45 =**1127.25 cm^3** (iii) Volume of ladoo= $\frac{4}{3}\pi r^3$ $=\frac{4}{3}\cdot\frac{22}{7}\cdot 1^3=\frac{\mathbf{88}}{\mathbf{21}}cm^3$	1 1 ½ ½ 1 ½ ½ ½ ½
38	(i) $x^2-24x+128$ $= x^2-16x-8x+128$ $= (x-8)(x-16)$ So zeros are **x=8 and x=16** (ii) $\alpha+2=3+2=5$ $\beta-1=-2-1=-3$ $P(x)=x^2-\big(5+(-3)\big)x+5(-3)=\boldsymbol{x^2-2x-15}$ OR $P(-3)=9k-18+12=9k-6=0$ ⇨ 9k=6 ⇨ **K=6/9=2/3** (iii) $(\alpha+\beta)^2=(8+16)^2=24^2=\mathbf{576}$	½ ½ ½ ½ 1 1 1 1